FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS — GRADE 10 | SUB-STRAND 1.2: INDICES AND LOGARITHMS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You will write a structured response to the driving question using everything you have learned across all 8 lessons in this unit. Read the driving question carefully, then complete each section below in order. Your goal is to show — clearly and with mathematical evidence — how indices and logarithms are powerful tools for real Kenyan professionals, not just classroom exercises. Use the Kericho tea farm phenomenon as your anchor throughout, but you may also draw on other Kenyan contexts you have explored (population growth in Nairobi, pH of Lake Nakuru, sound levels at a Kisumu concert, compound interest at a Mombasa bank, etc.). Write in your own words. Show all working where prompted. A model exemplar is provided after each prompt to guide your thinking — study it before you write your own response.

Section 1 — Restating the Problem: Why Ordinary Arithmetic Breaks Down	
In 3–5 sentences, explain WHY the Kericho tea farm manager's calculation becomes unwieldy using ordinary multiplication. What specifically makes large (or very small) numbers difficult to handle without a smarter mathematical tool? Reference at least one concrete number from the phenomenon in your answer.	When the Kericho tea farm manager tries to multiply numbers like 2^{10} (1 024 plants) by 2^3 (8 grams per plant) and then scale that up across an entire season's worth of harvest cycles, the digits pile up rapidly. A single multiplication of two six-digit numbers already requires keeping track of dozens of partial products, and one misplaced digit ruins the entire answer. The problem grows even worse when the numbers involve decimals or very small quantities — for example, the mass of a single tea-leaf cell measured in milligrams. Ordinary arithmetic gives us no shortcut: every extra zero in a number adds another layer of working. This is exactly the gap that index notation and logarithms were invented to close.

Section 2 — Index Notation and the Laws of Indices: The Language of Repeated Multiplication	
Explain what an index (exponent) means and state ALL the laws of indices you	An index tells us how many times a base number is multiplied by itself. Writing 2^8 means $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$; the base is 2 and the index is 8. This compact notation saves space and, more importantly, unlocks powerful rules.

have learned. For each law, write the general rule in symbolic form AND give a worked example set on the Kericho tea farm or another Kenyan context. Then answer: why does increasing the exponent by 1 (e.g., from 2^9 to 2^{10}) double the harvest? Show the arithmetic that proves it.	The Laws of Indices: 1. Product Law — $a^m \times a^n = a^{m+n}$ Example: The farm has 2^4 rows of plants and each row is extended by a factor of 2^3. Total plants = $2^4 \times 2^3 = 2^{4+3} = 2^7 = 128$. 2. Quotient Law — $a^m \div a^n = a^{m-n}$ Example: A warehouse stores 2^8 kg of tea; it ships 2^5 kg. Remaining = $2^8 \div 2^5 = 2^3 = 8$ kg. 3. Power Law — $(a^m)^n = a^{mn}$ Example: Each of 2^3 estates is subdivided into $(2^2)^3 = 2^6 = 64$ sub-plots. 4. Zero Index — $a^0 = 1$ Example: $5^0 = 1$; even if a Mombasa exporter raises any price base to the power zero, the multiplier is 1 — no change. 5. Negative Index — $a^{-n} = 1/a^n$ Example: A tea-leaf cell has a diameter of 2^{-3} mm = 1/8 mm = 0.125 mm. 6. Fractional Index — $a^{(1/n)} = \sqrt[n]{a}$ Example: The cube root of 2^6 kg of fertiliser = $2^{(6 \times 1/3)} = 2^2 = 4$ kg per equal pile. Why does adding 1 to the exponent double the harvest? $2^{10} \div 2^9 = 2^{10-9} = 2^1 = 2$. So $2^{10} = 2 \times 2^9$. Every time the exponent increases by 1, the value is multiplied by the base (2). This is the mathematics behind exponential growth — a concept Kenyan public-health scientists use to model disease spread and that Nairobi property investors use to project compound returns.

Section 3 — Logarithms: Turning Multiplication into Addition

Define a logarithm in plain language AND in formal notation. Explain the relationship between indices and logarithms (they are inverses). State the three main logarithm laws, give a worked example for each using a Kenyan context, and then walk through a complete four-figure table calculation — step by step — that solves a realistic multiplication problem a Kericho farm manager might actually face. Finally, explain in one paragraph WHY adding two small decimals from a log table gives the same answer as multiplying two large numbers.	A logarithm answers the question: 'What power must I raise the base to in order to get this number?' In formal notation: if $a^x = N$, then $\log_a N = x$. For example, $\log_2 8 = 3$ because $2^3 = 8$. Logarithms and indices are inverse operations — just as subtraction undoes addition, taking a logarithm undoes raising to a power. On a graph, $y = 10^x$ and $y = \log_{10} x$ are mirror images across the line $y = x$. The Three Laws of Logarithms (base 10, common logarithms): 1. Product Law — $\log(A \times B) = \log A + \log B$ Example: Total tea export = 3 400 kg × 2 700 kg-equivalent bales. log(3 400 × 2 700) = log 3 400 + log 2 700 = 3.5315 + 3.4314 = 6.9629 → antilog = 9 180 000. 2. Quotient Law — $\log(A \div B) = \log A - \log B$ Example: A Kisumu factory divides 85 600 litres of liquid fertiliser among 320 farms. log(85 600 ÷ 320) = log 85 600 − log 320 = 4.9325 − 2.5051 = 2.4274 → antilog = 267.5 litres each. 3. Power Law — $\log(A^n) = n \times \log A$ Example: A Nairobi bank offers 8% compound interest. After 12 years, KES 50 000 grows by a factor of $(1.08)^{12}$. $\log(1.08^{12}) = 12 \times \log(1.08) = 12 \times 0.0334 = 0.4008$ → antilog = 2.515. Amount = 50 000 × 2.515 = KES 125 750.

	Four-Figure Table Worked Example — Kericho Farm Production: Problem: A tea estate harvests 3 250 kg in the first cycle and expects production to scale by a factor of 4 180 due to new irrigation. Find total projected production. Step 1 — Find log 3 250: Look up 32 in the main table, column 5 → 0.5119. Add mean difference for the 0 → 0.5119. With characteristic: log 3 250 = 3.5119. Step 2 — Find log 4 180: Look up 41 in the main table, column 8 → 0.6212. Add mean difference for 0 → 0.6212. With characteristic: log 4 180 = 3.6212. Step 3 — Add: 3.5119 + 3.6212 = 7.1331. Step 4 — Find antilog 0.1331: Look up 0.13 in the antilog table, column 3 → 1.358. Add mean difference for 1 → 1.359. With characteristic 7: answer = 1.359×10^7 = 13 590 000 kg. Step 5 — State answer: The projected total production is approximately 13 590 000 kg. Why does this work? Every number can be expressed as a power of 10. When you look up log 3 250 = 3.5119, you are really saying 10^3.5119 = 3 250. When you look up log 4 180 = 3.6212, you are saying 10^3.6212 = 4 180. Multiplying those two numbers means multiplying 10^3.5119 × 10^3.6212 = 10^(3.5119 + 3.6212) by the product law of indices. So you only need to add the exponents — the two small decimals in the table — to find the exponent of the answer, then convert back with the antilog table. The log table is simply a pre-computed list of exponents of 10, and the logarithm laws are a direct consequence of the index laws.

Section 4 — Connecting Both Tools to Real Kenyan Contexts	
Choose TWO different real-world Kenyan contexts (other than the Kericho tea farm) in which indices OR logarithms are genuinely useful. For each context: (a) name the context and the professional who would use it, (b) write the mathematical problem they face, (c) show a full worked solution using either index laws or log laws, and (d) write one sentence connecting the mathematics back to the driving question.	Context 1 — Population Growth in Nairobi (Urban Planner / Demographer) (a) A Nairobi city planner needs to project the population of a new satellite town that currently has 12 500 residents and is growing at 6% per year. (b) Problem: What will the population be after 20 years? Model: $P = 12\,500 \times (1.06)^{20}$. (c) Solution using logarithms: log P = log 12 500 + 20 × log 1.06 log 12 500 = 4.0969 log 1.06 = 0.0253 20 × 0.0253 = 0.5060 log P = 4.0969 + 0.5060 = 4.6029 P = antilog 4.6029 = $4.007 \times 10^4 \approx$ 40 070 residents. (d) Just as the Kericho farmer used logarithms to handle large harvest figures, a Nairobi planner uses the same technique to multiply a growth factor raised to a large power — turning what would be 20 successive multiplications into a single addition. Context 2 — Acidity of Lake Nakuru (Environmental Chemist / Kenya Wildlife Service Scientist) (a) A KWS chemist monitoring flamingo habitats at Lake Nakuru measures hydrogen ion concentration $[H^+] = 2.5 \times 10^{-9}$ mol/L and needs to report the pH. (b) Problem: Calculate the pH, defined as $pH = -\log_{10}[H^+]$. (c) Solution: $pH = -\log(2.5 \times 10^{-9})$ $= -(\log 2.5 + \log 10^{-9})$

	$= -(0.3979 + (-9))$ $= -(0.3979 - 9)$ $= -(-8.6021)$ = 8.60 (alkaline, which supports flamingo feeding). (d) Here the logarithm compresses an unmanageably small number (0.0000000025) into the tidy, human-readable value 8.60 — exactly the kind of 'making sense of numbers too small for ordinary arithmetic' that the driving question asks about.

Section 5 — Answering the Driving Question Directly	
Write a full, coherent paragraph (minimum 8 sentences) that directly and completely answers the driving question. Your answer must: (1) explain what indices do, (2) explain what logarithms do, (3) state how they are related to each other, (4) name at least THREE different Kenyan professionals who benefit from these tools, (5) reference the Kericho tea farm phenomenon explicitly, and (6) include at least one specific numerical example. This is your 'capstone' paragraph — write it as if you are explaining to a fellow student who missed the entire unit.	The Kericho tea farmer's problem — calculating the total harvest when numbers grow into the millions — is not unique to farming; it is a challenge faced by scientists, engineers, and business people all across Kenya whenever they work with quantities that grow or shrink exponentially. Index notation solves the first part of the problem by giving us a compact language for repeated multiplication: instead of writing 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2, we write 2^{10}, and the laws of indices let us multiply, divide, and raise these compact expressions to further powers with simple arithmetic on the exponents alone. This is why the farm manager sees the harvest double every time the exponent increases by just 1 — because $2^{n+1} = 2 \times 2^n$, and the index law makes that relationship visible and calculable. Logarithms extend this power even further: because every positive number can be expressed as a power of 10, we can represent any multiplication problem as an addition of exponents, look those exponents up in a four-figure table, add two small decimals, and then convert back — arriving at the same answer as a long, error-prone multiplication in a fraction of the time. Logarithms and indices are inverse operations, two sides of the same mathematical coin, and mastering one deepens your understanding of the other. A Kericho tea estate manager uses log tables to compute seasonal production totals in minutes; a Nairobi demographer uses the power law of logarithms to project a city's population twenty years into the future from a single growth rate; a Kenya Wildlife Service chemist at Lake Nakuru uses the definition of pH — itself a logarithm — to translate a hydrogen ion concentration of 0.0000000025 mol/L into the readable value 8.60. In every case, the mathematics of indices and logarithms does the same job: it tames numbers that are too large or too small for ordinary arithmetic by translating multiplication into addition, and enormous or microscopic quantities into compact, manageable expressions. The 'magic' the farmer's colleague demonstrated — adding two small table values to get the answer to a six-digit multiplication — is not magic at all; it is the product law of logarithms, which is itself a direct consequence of the product law of indices, applied through a table of pre-computed powers of 10. Once you understand this connection, you hold a tool that has been used by Kenyan farmers, scientists, and engineers long before electronic calculators existed — and that continues to illuminate the deep structure of exponential relationships in our world today.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Index Notation and Laws of Indices	Correctly states and applies all six laws of indices (product, quotient, power, zero, negative, fractional) with accurate symbolic notation and fully worked, contextually relevant Kenyan examples for each. Clearly explains why increasing the exponent by 1 doubles the value, with algebraic proof.	Correctly states and applies at least four laws of indices with mostly accurate worked examples. Explains the doubling effect with some supporting arithmetic, though the algebraic justification may be incomplete.	States fewer than four laws of indices or makes consistent errors in applying them. Worked examples are missing, incomplete, or contain arithmetic mistakes. The explanation of doubling is absent or incorrect.

Understanding of Logarithms and Their Relationship to Indices	Provides a precise plain-language and formal definition of a logarithm. Clearly and accurately explains that logarithms and indices are inverse operations. States all three log laws correctly with fully worked Kenyan-context examples. Explains convincingly, with reference to index laws, why log tables convert multiplication to addition.	Provides an adequate definition of a logarithm and notes the inverse relationship with indices. States at least two log laws correctly with mostly accurate examples. Gives a reasonable explanation of the multiplication-to-addition conversion, though it may lack full mathematical depth.	Definition of logarithm is vague or incorrect. The inverse relationship with indices is not mentioned or is confused. Fewer than two log laws are stated, or examples contain significant errors. The explanation of why log tables work is absent or fundamentally incorrect.
Four-Figure Table Calculation (Procedure and Accuracy)	Demonstrates a complete, correctly sequenced four-figure log table calculation: correctly identifies characteristics and mantissas, accurately reads mean differences, adds logarithms, and correctly uses the antilog table to state a final answer with appropriate units and rounding.	Demonstrates most steps of the four-figure table procedure correctly. May make a minor error in reading mean differences or in stating the final antilog, but the overall method is clear and the answer is approximately correct.	The four-figure table procedure is incomplete or contains multiple errors in identifying characteristics, reading the table, or finding the antilog. The final answer is significantly incorrect or the method cannot be followed.
Application to Real Kenyan Contexts	Selects two distinct, realistic Kenyan professional contexts (beyond the tea farm phenomenon). Each context includes a clearly stated problem, a fully worked solution using index or log laws, correct numerical answers, and an explicit sentence linking back to the driving question.	Selects two Kenyan contexts with adequately stated problems and mostly correct worked solutions. The link to the driving question is present in at least one context, though it may be brief or implicit.	Fewer than two Kenyan contexts are addressed, or the contexts chosen are not realistic/relevant. Worked solutions are incomplete or incorrect. No explicit connection is made back to the driving question.
Driving Question Response (Coherence, Completeness, and Mathematical Accuracy)	The capstone paragraph directly and completely answers the driving question in at least 8 sentences. All six required elements are present (indices explained, logarithms explained, inverse relationship stated, three Kenyan professionals named, tea farm phenomenon referenced, specific numerical example included). Mathematical statements are accurate and the writing is clear and logically sequenced.	The capstone paragraph answers the driving question adequately and contains at least 6 sentences. At least four of the six required elements are present. Mathematical statements are mostly accurate with minor omissions or imprecision in language.	The capstone paragraph is fewer than 6 sentences, is vague, or does not directly address the driving question. Fewer than three required elements are present. Mathematical statements contain errors or the response reads as a list of definitions rather than a coherent answer.